

Improving The Postal Museum's Online Image Licensing and Distribution Process

An Interactive Qualifying Project
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This report represents the work of WPI undergraduate students submitted to the faculty as evidence of completion of a degree requirement. WPI routinely publishes these reports on its website without editorial or peer review. For more information about the projects program at WPI, please see <https://www.wpi.edu/project-based-learning>.

Abstract

Digital image licensing has become a key revenue stream for many museums, including The Postal Museum (TPM) in London. TPM sought to improve internal efficiency and the public's ease of use of their licensing process, which runs primarily on the image platform PhotoShelter. We benchmarked TPM's pricing and licensing practices against other museums, solicited employees' suggestions, investigated online systems, and used this information to develop an improvised image licensing process. In the end, we proposed a new pricing structure, researched and created a plan for publicizing TPM's PhotoShelter portal, and developed additional functionality – an image quality analysis software and a user guide for clients.

Acknowledgements

The project team would like to express our sincere gratitude to The Postal Museum for the opportunity to work on this project and contribute to the continued development of its image licensing and distribution process.

We are especially grateful to Louis Porter, Chris Taft, and Gavin McGuffie for their guidance, expertise, and continued support throughout the project. Their willingness to share their knowledge, provide feedback, and discuss the Museum's collections, workflows, and goals was invaluable to the success of this work.

We would also like to thank Professor Dominic Golding and Professor Laureen Elgert for their advice, encouragement, and mentorship throughout the project. Their guidance helped shape both the direction of our research and the development of our final recommendations.

We would like to thank all members of The Postal Museum staff who contributed their time, insights, and perspectives during discussions and meetings. Their cooperation and engagement were essential to the completion of this project.

Finally, we extend our appreciation to everyone who contributed their time and knowledge through interviews, discussions, and reviews. Their perspectives were essential in helping us better understand the challenges and opportunities associated with image distribution and licensing at The Postal Museum.

Executive Summary

Amid reduced UK government funding, The Postal Museum (TPM) is among many museums seeking to diversify revenue sources. As public funding becomes less reliable, institutions increasingly depend on internally generated income such as image licensing, retail, and memberships (Aroles et al., 2022). TPM seeks to increase revenue by improving its image commercialization process, but the established system for handling image requests is inefficient and labor-intensive. Requests are managed manually, involving several back-and-forth communications, individualized pricing, and significant staff involvement. This slows processing, increases operational costs, and limits potential revenue. It can also discourage potential clients due to delays and complexity, further restricting growth. Addressing these inefficiencies is essential for TPM to better leverage its digital assets and strengthen their long-term financial sustainability.

Methods/ Approach

At the start of this project, The Postal Museum already possessed many of the components necessary for an effective image distribution system, including a substantial digital archive, an established licensing process, staff expertise, and the use of PhotoShelter as a digital asset management platform. However, the existing process relied heavily on manual communication, individualized pricing decisions, and case-by-case copyright review, creating inefficiencies and increasing staff workload. The overall goal of this project was to help The Postal Museum better fulfill their mission by improving their online image distribution process. We achieved this goal through five objectives:

- 1) Objective 1: We suggested a revised and balanced image pricing structure informed by internal evaluation and external benchmarking.
- 2) Objective 2: We assessed the functionality and usage of DAMS within TPM, such as the online catalogue and PhotoShelter, and identified employees' desired improvements to the current image licensing process.

- 3) Objective 3: We prototyped changes to PhotoShelter, built digital tools, and developed strategies to improve the image licensing process.
- 4) Objective 4: We presented a proof-of-concept system along with a potential implementation plan for stakeholders.

Key Findings

TPM's current image request process is highly manual and relies heavily on staff intervention at each stage. TPM's online catalogue is currently the typical entry point for customers; it is the platform for them to find desired images before making a request. When a client submits an image inquiry, museum staff must search PhotoShelter on the client's behalf to identify suitable images, assess image quality, determine copyright status and licensing restrictions, negotiate a fee based on the intended use, and coordinate payment and invoicing with the finance team. Our investigation found that this is both time-consuming and inefficient, requiring significant staff effort and numerous email exchanges with clients to clarify requirements, confirm pricing, and arrange payment. Through discussions with TPM, we witnessed exactly how tedious this process can get. The complexity of individual image assessment and the non-automated pricing structure further increase administrative workload, as each request often requires individual judgement rather than following a standardized process. Several opportunities for improvement were identified, including:

- Implementing image flagging or tagging systems to simplify rights management
- Reducing the number of manual invoices generated
- Minimizing back-and-forth email communication through greater automation and self-service functionality
- Introducing a clearer, and more structured pricing model, utilizing PhotoShelter's individual tiers.

An evaluation of The Postal Museum's existing image pricing practices, combined with benchmarking against comparable museums and cultural institutions, revealed that the current pricing approach lacked consistency and was difficult to scale within an automated system. Pricing decisions were often made on a case-by-case basis, requiring staff judgement and limiting opportunities for self-service purchasing. Benchmarking demonstrated that many

institutions use structured pricing tiers based on factors such as image resolution, intended use, and commercial purposes. These findings informed the development of a revised pricing structure that balances revenue generation with accessibility while supporting implementation through PhotoShelter's e-commerce functionality. The proposed model provides greater transparency for clients, reduces administrative burden for staff, and creates a more scalable framework for future image distribution.

Our review found that PhotoShelter enables users to search for assets using keywords, phrases, and filenames, supported by filters such as file type, orientation, date, and other attributes. The platform also supports custom metadata, including the ability to distinguish between free and paid images. Through the portal, users can discover, request access to, and download content, making it suitable for both internal and external audiences (PhotoShelter, 2024).

The evaluation identified several strengths. PhotoShelter was found to be user-friendly, highly customizable, and effective at protecting digital assets. It also offered strong integration capabilities with external platforms, including e-commerce systems, which aligned with TPM's objectives. Furthermore, the platform appeared adaptable to a wide range of sectors and user groups beyond photography, including government and healthcare organizations, which suits how the museum would be using it.

The evaluation also identified several limitations. PhotoShelter represents a significant financial investment, and some organizational requirements may require additional tools or integrations to be fully addressed. Since TPM already had the existing infrastructure for this program, this was less of a concern for our project. Due to this, we identified areas where supplementary systems may be beneficial, particularly in relation to e-commerce functionality.

After setting up the public portal, we then turned to improve the user experience of both the TPM staff and customers using the PhotoShelter portal by streamlining the image distribution process. We first identified several grievances from the TPM staff about the system and conducted investigations on our own to find potential problems. Three major concerns were identified – **image quality evaluation, image searchability, and information availability.**

Image Quality Evaluation: TPM staff want to license only images that meet certain quality standards, but they cannot easily evaluate the quality of images in the library. This is complicated by the fact that images cannot be zoomed into natively without downloading, and there are roughly 50,000 images stored. Therefore, we suggested using laptop trackpad to zoom in and requested a zoom-in feature from PhotoShelter support. We also developed an image quality analysis software capable to scan the entire library and tag images with potential quality issues.

Image Searchability: An investigation revealed the online catalogue to be a viable starting point of the licensing process, but we also discovered the inconsistency between the metadata of some images in the online catalogue and in the PhotoShelter library. This makes it difficult or impossible for the customers to find the corresponding image in the library using the information in the catalogue. After many inquiries, we discovered three PhotoShelter features that can assist users in searching for images in PhotoShelter without detailed manual metadata.

Information Availability: There were frequent trivial exchanges between the staff and the customers due to a lack of public information about the image licensing process. Therefore, we wrote a user guide page that outlines the entire process of purchasing images and answers the frequently asked questions. We hope by making this information available to the public, it will streamline the communication flow between staff and customers. Ideally, customers will be able to largely guide themselves through the image purchasing process.

Recommendations and Conclusion

Our project resulted in a proof-of-concept system that demonstrates how The Postal Museum could improve access to its image collection while reducing administrative burden on staff. We created the Image Quality Examiner (IQE), a program written in python that connects to PhotoShelter's API and analyzes the metadata of images to identify images that don't meet TPM's criteria. We accompanied TPM with the user guide as well as an implementation plan, which contains findings from our research as well as future plans to maximize efficacy of the image licensing process. Some of these future recommendations include:

- Manually correcting metadata inconsistencies between the online catalogue and PhotoShelter as they are identified

- Directing customers to the online catalogue initially to find basic image information prior to accessing PhotoShelter
- Publicizing the user guide for enhanced customer self-service
- Streamlining the image pricing structure by reducing the number of pricing tiers for each item
- Increasing licensing prices for single tv/film purposes and for personal use
- Using the IQE tool to remove or recreate substandard images identified in PhotoShelter

These recommendations provide a practical foundation for future development. By building upon the work completed during this project, The Postal Museum can enhance the management of its digital archive, optimize image monetization practices, and better fulfill its mission through a more sustainable digital infrastructure.

Authorship

For this report, all group members contributed to the writing process, with each member taking primary responsibility for specific sections. The individual contributions were subsequently integrated into a single document and edited primarily by another member, before being reviewed and approved by the rest of the group.

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1. Introduction

A museum's collection gains its true value not from the objects themselves, but from the stories and names tied to them. Every artifact reflects aspects of global history, which provides cultural significance. This value is demonstrated not only in public exhibitions but also in the financial systems that allow museums to operate. One way museums translate this cultural value into revenue is through the commercialization of copyrighted materials in their collections, granting licenses and access to individuals who wish to use these materials in their own work. As museums expand their digital presence, this practice has become increasingly important, enabling wider audience outreach while supporting sustainable income. Research on cultural heritage digitization and museum economics shows that digital access and licensing models are now key to modern museum sustainability strategies, particularly as institutions balance public access with financial resilience (Mohammed & Metwaly, 2024). One example is The Postal Museum (TPM) in London, which generates revenue by commercializing its image collection. This archive contains thousands of historical images documenting centuries of British postal history, making it both culturally and financially significant.

Beyond its cultural importance, TPM's image archive functions as a structured digital asset that supports institutional operations and outreach. However, its effectiveness depends on how efficiently it can be accessed, managed, and licensed. Many museums now use Digital Asset Management Systems (DAMS) to organize, store, and distribute media in a scalable way. Despite these advances, implementation and workflow efficiency vary widely across institutions, and legacy or manual processes can still limit the potential of digital collections. This gap between technological capability and operational practice directly impacts how effective the revenue generated from digital assets can be.

Amid reduced UK government funding, TPM is among many museums seeking to diversify revenue sources. As public funding becomes less reliable, institutions increasingly depend on internally generated income such as image licensing, retail, and memberships (Aroles et al., 2022). TPM seeks to increase revenue by improving its image commercialization process. However, the established system for handling image requests is inefficient and labor-intensive.

Requests are managed manually, involving several back-and-forth communications, individualized pricing, and significant staff involvement. This slows processing, increases operational costs, and limits potential revenue. It can also discourage potential clients due to delays and complexity, further restricting growth. Addressing these inefficiencies is essential for TPM to better leverage its digital assets and strengthen their long-term financial sustainability. Overall, this project worked to strengthen TPM's ability to fulfil its mission through improved digital image distribution. By enhancing the management of its digital archive and optimizing image monetization practices, this project supports a more sustainable digital infrastructure for the museum.

Our goal was to help the Postal Museum (TPM) better fulfill their mission by improving the online image distribution process. The idea was to improve how TPM manages its digital archive and the monetization process involved with image distribution. We have addressed this through four objectives:

- 1) We suggested a revised and balanced image pricing structure informed by internal evaluation and external benchmarking.
- 2) We assessed the functionality and usage of DAMS within TPM, such as the online catalogue and PhotoShelter, and identified employees' desired improvements to the current image licensing process.
- 3) We prototyped changes to PhotoShelter, built digital tools, and developed strategies to improve the image licensing process.
- 4) We presented a proof-of-concept system along with a potential implementation plan for stakeholders.

2. Background

In this section, we aim to better contextualize our project and the motivations behind it. We will start with the history of museums and their importance to society, then move into the organization of museum collections and what roles archives serve in the context of a museum. Then, we will provide a comprehensive breakdown of the current financial situation of museums and their journey to diversify income sources. Afterwards, we will introduce The Postal Museum (TPM), examining its history and mission, as well as the history and current methods of its image

licensing business. Lastly, we examine the role of copyright in museums, and its role in TPM's image licensing practice.

As defined by The International Council of Museums (ICOM), a museum is “a not-for-profit, permanent institution in the service of society that researches, collects, conserves, interprets and exhibits tangible and intangible heritage. Open to the public, accessible and inclusive, museums foster diversity and sustainability. They operate and communicate ethically, professionally and with the participation of communities, offering varied experiences for education, enjoyment, reflection and knowledge sharing” (ICOM, 2022). It is primarily through the growth, maintenance, interpretation, and display of their collections that museums serve society. Museums create thematic exhibits, which are curated collections of their artifacts, so that visitors can explore their collections firsthand. Exhibits are displayed in galleries within museums and can be permanent or temporary. Exhibits also often travel between institutions, or between different galleries in a museum. Typically, 10% or less of a museum's collection is displayed in permanent exhibits on average; the rest is either used to create temporary exhibits, lent to other institutions for research and/or display, or kept in storage at the museum or off site (“Stored Collections of Museums,” 2024).

To accompany their collections, museums dedicate resources towards conducting and promoting research, as well as towards the education and engagement of the public. Museums research their own collections to establish better interpretations of their collections, enabling them to curate more accurate exhibits and better serve their mission. To support research, they frequently offer researchers and students access to their full collection, including access to the archive. A museum's archive is a specialized collection that preserves thousands of primary sources, records, graphics, and other pieces of recorded data for documenting the history of the museum and the context behind the artifacts in its collection (Society of American Archivists, 2016). The vast majority of artifacts that are not publicly displayed in a museum are stored in the archive. Museums also support public education through community outreach events, specialized exhibits, educator training programs, and many other initiatives. In the United States alone, museums spend over \$2 billion annually on pre-collegiate education activities (“Museums and P-12 Education,” 2017).

2.1. Museums Seek to Diversify Funding Sources

Despite their grand goals and service to society, museums still need funding to function. Employees must be paid, buildings must be maintained, items must be preserved, and new items for the collection must be purchased. The items in the museum must also be restored to their glory, as well as preserved so people may see them for as long as possible. All these factors can add up to quite a large sum, and securing adequate funding is a constant struggle for many, if not most museums in the UK. In general, British museums get funding from admission, retail and catering sales, grants, donations, property rent, sponsorships, local authorities support, and charity support, as well as any unique, museum-specific sources (Li, 2024).

In the UK, a significant factor in the struggle for funding is that local authorities' (LA) support has been decreasing steadily. For example, the local authority expenditure on cultural activities dropped by 42% (adjusted for inflation) between 2009 and 2023 (Figure 1) with a slight increase in 2021-2022 to encourage tourism after the COVID 19 pandemic, as shown in Figure 1.

Figure 3.4: Net Current Expenditure on Total Cultural and Related Services from 2009-10 to 2022-23 by Local Authorities (£millions)

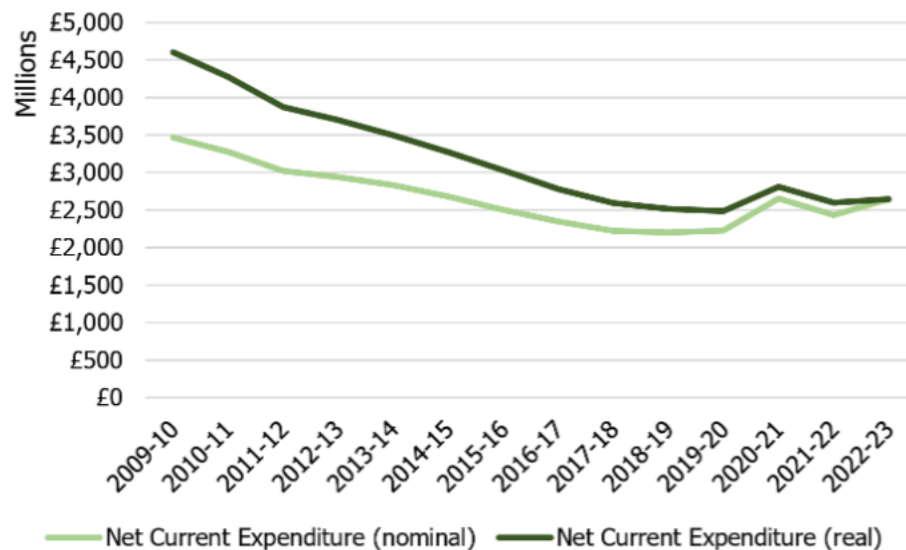


Figure 1: Graph of the expenditure on cultural activities from 2009 to 2023 (Research to Understand the Levels of Public Investment in Museums, n.d.)

This is due to the UK utilizing austerity cuts to ensure that government budgets are followed without reducing their expenditure in areas deemed of higher importance than others. An independent report from the Esmée Fairbairn Foundation noted from interviews with

anonymous stakeholders that LAs are generally reducing cultural activity funding in favor of more immediate societal needs, like social care and homelessness services (Local Authority Funding Report, n.d.). The LAs also had their total funding reduced, which forces even more funding to be siphoned away from museum investments (Rex & Campbell, 2021). A more recent positive event affecting museums is a 1.5-billion-pound investment into museums by the UK central government over 5 years (DCMS Announces £1.5bn to Tackle Underfunding across Culture Sector - Museums Association, 2026). The money received will help struggling and up-and-coming museums significantly in efforts towards becoming a self-sustaining business but are far from enough to balance out the years of diminishing LA support. The erratic nature of LA and central government regarding funding is simply not a reliable source of income for museums.

When the COVID-19 pandemic occurred, the delicate balance museums had built by balancing expenses with the decreasing LA financial support fell apart. Since the pandemic, the UK National Audit Office (NAO) has a 13% reduction in total visitation among the fifteen museums they assessed. There was also an increase in general competitiveness between museums for visitor time (Museums and Galleries Boost Self-Generated Income but Face Risks to Their Finances - NAO Press Release, 2026). The decrease in visitors has a ripple effect, decreasing nearly every single income source a museum has, and a higher competitive market means that if museums wish to improve their revenue through their current sources, they must increase their appeal to visitors as much as possible. They may need to use advertisements, sales, or events to bolster their appeal, often at a significant expense to the museum at an uncertain level of return. In contrast, diversifying a museum's methods of revenue, especially towards ones the museum is already well-equipped for, can provide relatively low risk, low-cost streams of income that will enable the museum to improve the more important and lucrative funding sources. The Postal Museum receives minimal funding from LAs compared to many other museums in the UK. It is instead funded by the Postal Heritage Trust, ticket sales, retail, venue hire, donations, and grants, and is still exploring ways to diversify its revenue sources.

2.2. The Postal Museum

The Postal Museum (TPM) is a British postal communication theme museum located in London, United Kingdom. TPM was opened in 2017, and its archive dates back more than five

hundred years (Our History, TPM). The British postal network originally only served the royal court. King Charles I opened the network to the public in 1635, and the General Post Office was established under Oliver Cromwell and then King Charles II in 1660. Starting in the 1800s, organizing and preserving records became important duties of the post office. Sir Francis Freeling, Secretary of the Post Office from 1797 to 1836, introduced the system for recording minutes and reports for the first time. In 1838, the post office was mandated to maintain a public record by the first Public Records Act. Since the act, an archive began growing, and the record room of the General Post Office became the earliest predecessor of TPM. In 1969, Queen Elizabeth II opened the National Postal Museum in the City of London. Unfortunately, the museum was short-lived and closed in 1998 when its building was sold. Although the museum was closed, its collections were retained by the post office that eventually became Royal Mail. In 2004, the collections were transferred by Royal Mail to a newly formed independent charity called the Postal Heritage Trust. The physical building of TPM fully opened on September 4, 2017. From that time onward, TPM has been the public face of the charity.

TPM's mission is to "tell the story of postal communication and its impact on a global society" (About us, TPM). It wants to use the story to inspire people and bring positivity into their lives. Its five core aims/visions are:

- Create welcoming experiences for anybody to connect with the museum and each other.
- Collaborate creatively to deliver content that meets our audience's needs.
- Look after, develop and be curious about our collection to ensure the story of postal communication is relevant, inspiring and inclusive.
- Minimize the environmental impact of our activities and communicate effectively about our environmental responsibility.
- Develop a sustainable and dynamic business model to enable our work as a charity.

Like other museums, TPM also has three main responsibilities – preserve the collection and archive, conduct and promote research, and education. Preservation of their collection is TPM's third core vision. Collection and archive have different purposes and emphasis on the type of items to preserve (Schmidt, 2011). Museum collections are not limited to items on display; they encompass all objects held by the museum for preservation, research, interpretation, and exhibition. This can result in items in the collection being three-dimensional artifacts, and they may also include archival materials and other object types. Museums typically

maintain diverse collections to support a range of educational, research, and exhibition activities and to engage a broad audience. TPM's collection contains postal theme items like old vehicles, uniforms, and pillar boxes. Some of the items are shown below in Figure 2.



Figure 2 (Left): Mail Guard's Frockcoat. Manufacturer: Herbert & Co, London 1875-1882.

Figure 3 (Right): A post box for Air Mail. Now a part of the collection of the British Postal Museum & Archive. Displayed at Postal Museum (London). Image taken by Gaius Cornelius.

On the other hand, archives focus on preserving historical documents for future reference and research (Schmidt, 2011). They typically center around a certain person or organization, and access is more restricted, usually requiring identification and an appointment to access materials. TPM's archive is officially under the name of Royal Mail Archive, but TPM is the one managing it. Its archive mostly contains post office records and minutes (Royal Mail Archive Overview, 2025). In total, TPM has over 60,000 objects in the collection and thousands of documents in the archive (Collections, TPM). To ensure their collection keeps existing in good condition, TPM has a Conservation Studio to preserve and conserve its collection (Collection Care, TPM). The studio has extensive experience in conserving paper documents and vows to keep their skills and knowledge up to date (Conservation Studio, TPM). In terms of TPM's collecting practices, the museum "... will ensure that both acquisition and disposal are carried out openly and with transparency" (Collections Development Policy, 2021). Collection accessions and deaccessions follow strict guidelines. TPM only acquires an item for the collection when there is a sound curatorial reason; the act of acquisition does not damage the item significantly, and the original owner does not object. Occasionally, the TPM may deaccession an artifact in the collection through its sale, donation, destruction, or return to the original owner.

TPM encourages researchers and members of the public to research the archives. People may search the online catalogue (catalogue.postalmuseum.org), a resource where individuals can browse through tens of thousands of item records that the Museum's collection encompasses, to explore their postal service family history, study original material, or browse the reference library (The Archive, TPM). The catalogue can also be used to book an in-person visit to the museum to browse the archives. TPM uses the archive and larger collection for its own internal research in developing exhibits and programs, and staff may also conduct and collaborate in academic research. For example, TPM is a partner in a three-year project exploring the timing and geography of ill health and the responses of the Post Office and its workforce.

2.3. Image Licensing Business of TPM

One way that TPM diversifies their revenue is through image-licensing. Operating in an increasingly digital environment, and amidst COVID-19 business pressures, many museums including TPM have undergone a period of digitalization to increase internal efficiency, reach a wider audience, and maintain a modern public image. The process of digitalization in museums encompasses the integration of digital technologies into their core business operations. These changes include managing collections, establishing / updating websites, e-commerce integration, and collecting data for analytics. Another key change to museums brought on by digitalization is the digitization of their archives. Digitization is the process of converting physical documents, records, books, and artifacts into a digital format (Xia et al., 2024). For physical artifacts, this is often done by a professional who photographs the item such that the nature of the artifact is preserved. Digitized items carry several advantages, as they can be quickly accessed from a database and distributed online, along with associated metadata such as the item's date, format, and subjects, without the need to handle the original item. This improves both the accessibility and long-term preservation of items held in museum archives (Xia et al., 2024).

TPM generates revenue from images in its archive by selling licenses to clients seeking to use the Museum's images for personal interest. The Museum collects an image supply fee for all licensed images. Many items in TPM's archive are subject to copyright laws, in which case the client may need to pay an additional copyright licensing fee to TPM or the respective copyright holder to use the image. If the desired item has not yet been digitized, an additional digitization fee will be charged for the creation of a new image. TPM has a legal obligation to make other

documents in the archive accessible to the public, such as Royal Mail's operational reports or certain items donated to the Museum. These are made available without charge. TPM commercializes their images through an enquiry-based system, which involves the reception of enquiries from a client to purchase an image, and an employee from TPM working with the client to deliver an appropriate image at an agreed price. This price is determined considering the magnitude, format, and intended use of the image(s).

2.4. Copyright

Museums occupy a unique position when it comes to copyright. They exist as both archivists of others' creative work, while also being producers of their own. While seeking to support themselves, museums are increasingly seeking to gain revenue from both. For an institution like The Postal Museum, navigating copyright laws is not straightforward. Understanding the basis of what the Museum can legally charge for is foundational for this project.

The starting point for Copyright is the Designs and Patents Act 1988 (CPDA). This is the principal legislation governing intellectual property in the United Kingdom. The CPDA establishes that copyright protection applies automatically to creative works (photographs, artistic works, films and literary works) and that this protection lasts for various timeframes but generally seventy years after the creator's death (Copyright Act, 2021). Importantly, the law draws a firm line between owning a physical object and owning the copyright to it. In the words of The Postal Museum's internal documentation, "Just because an image is owned by us or is an accessioned part of one of the collections, this has no bearing on copyright" (Taft, 2021). For example, an Edwardian postcard may be in the Museum's collection; however, the reproduction rights are an entirely separate entity.

The Postal Museum's collection does not fall into one copyright category, and so there is no single approach to image licensing. The Museum images can be divided into three categories, each with their own legal implications.

The most straightforward category is content that the Museum holds the copyright directly. These are items like photographs taken by Museum staff, images created specifically for the Museum under commission, and original works produced in-house. If the photographer is an employee of the institution, the copyright automatically belongs to the institution (Taft, 2021).

The second category pertains to material that is held in the Royal Mail Archives. The Postal Museum manages the licensing of this archive on behalf of the Royal Mail Group. This means that the museum does not own the copyright; the Royal Mail Group does. The Museum's licensing agreement permits it to grant commercial licenses on behalf of the Group, within the boundaries of that agreement (Taft, 2021).

The final and most complex category is that of the items in the collection where copyright status is unclear, expired, or held by an unknown third party. For historic items determining copyright requires tracing both the creator and the date of their death, a process that is not always possible. The CDPA provides some guidance here: where a creator is unknown, copyright expires seventy years after the date of creation (Copyright Act, 2021). The National Archives further clarifies that published Crown Copyright material carries a fifty-year term of protection from the date of publication; unpublished Crown Copyright material can remain protected until as late as 2040 (National Archives, n.d.).

For a collection as historically deep as The Postal Museum's, these distinctions matter considerably. The Museum's own documents acknowledge that in some cases copyright simply cannot be ascertained with certainty and requires a risk-based judgement call rather than a definitive answer. The Postal Museum has developed an internal decision-making framework for assessing the copyright status of individual items. Figure 4 outlines the decision-making process that TPM goes through each time a license request is made. It begins by asking whether the creator of a work is known, then moves through factors such as the creator's date of death, the date of creation, and whether a publisher or rights holder can be identified. Works are generally treated as out of copyright when the creator died more than seventy years ago or, in some cases, when an older work has no traceable creator. Where the creator died more recently, or where a newer work has no identifiable rights holder, the work is treated as still in copyright or as having an untraceable copyright holder.

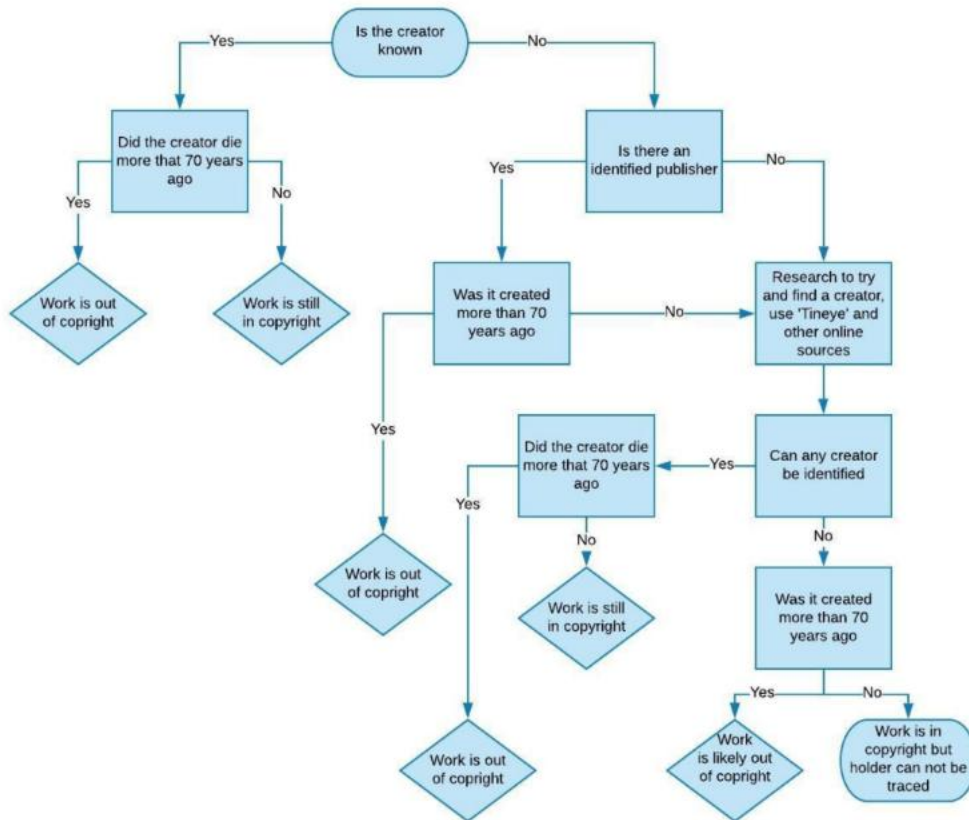


Figure 4: Process of museum when dealing with copyright (Taft, 2021)

A significant development in how UK copyright law applies to museum collections came with the 2023 court ruling in *THJ v Sheridan*. Though the case concerned financial software rather than museums, it clarified the standard a work must meet to qualify for copyright protection in the UK. For most of the twentieth century, courts applied a "skill and labour" test, which is if producing a work required meaningful effort, then it was generally considered original enough to attract protection (Rahmatian, 2024). *THJ v Sheridan* confirmed that copyright now requires the author to have made genuinely free and creative choices that stamp the work with their personal touch. This makes it unlikely a UK court would uphold a museum copyright claim over a faithful photographic reproduction of an out-of-copyright work, since the goal of such a photograph leaves little room for creative freedom. Crucially, even where copyright cannot be asserted, the Museum retains the ability to charge access fees and enforce usage restrictions through contract law (Korn & Waterman, 2024).

2.5 Conclusion

TPM's image licensing process depends on several connected systems: collection access, staff workflow, pricing, copyright review, client communication, and digital infrastructure. The Museum already has a valuable image archive and an established licensing process, but much of that process still relies on manual communication and case-by-case decision-making. Improving image distribution therefore requires an approach that considers both the technical system and the institutional judgment behind it. The following methods explain how our team will evaluate TPM's current process, gather input from staff and clients, compare possible tools, and develop a prototype for a more efficient and accessible image licensing system.

3. Methods:

Our goal was to help the Postal Museum (TPM) better fulfill their mission by improving the online image distribution process. The idea was to improve how TPM manages its digital archive and the monetization process involved with image distribution. We have addressed this through four objectives:

- 1) We suggested a revised and balanced image pricing structure informed by internal evaluation and external benchmarking.
- 2) We assessed the functionality and usage of DAMS within TPM, such as the online catalogue and PhotoShelter, and identified employees' desired improvements to the current image licensing process.
- 3) We prototyped changes to PhotoShelter, built digital tools, and developed strategies to improve the image licensing process.
- 4) We presented a proof-of-concept system along with a potential implementation plan for stakeholders.

Figure 5 breaks down our four main objectives into their related tasks. These objectives started from research-based work, such as reviewing image pricing, assessing digital asset management tools, and investigating desired functionality. As the project progressed, the focus shifted to design, which includes the configuration, testing, and presentation of the image distribution portal along the more streamlined process.

This chapter introduces the project scope, outlines the methods used for each objective, and explains how these components collectively contribute to our goal of improving TPM’s image distribution.

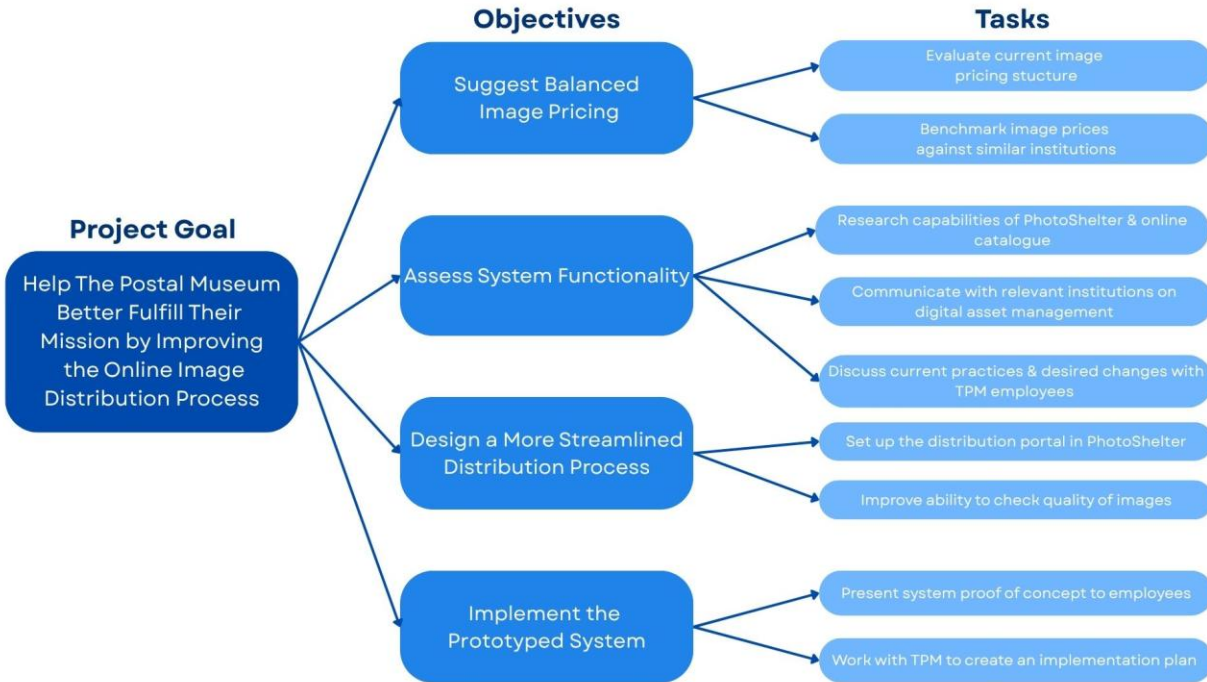


Figure 5: Project goal, objectives, and tasks outline diagram

3.1. Objective 1 – Suggest New Balanced Image Pricing

The Postal Museum’s image pricing structure has not changed since 2019 and needed updating to reflect institutional needs and copyright considerations. We evaluated the current image pricing structure for any immediate flaws or limitations that had arisen since the last update. We reviewed similar pricing schemes at other museums and archives by reviewing the information available on their respective websites supplemented by interviews with selected museums staff. This review allowed us to benchmark market rates and costs. We discussed our suggested pricing strategy with relevant TPM staff to avoid infringing any copyright laws that have been established since 2019.

3.1.1. Review Postal Museum Pricing

To properly evaluate the current image pricing structure at the Postal Museum, we analyzed the current pricing structure, as well as interviewed staff that had a hand in creating the

pricing structure, such as Louis Porter, a photographer for the museum, as well as Chris Taft, Head of Collections, Rose Brown, Archivist (Cataloguing), Helen Dafter, Archivist, and Gavin McGuffie, Senior Archivist. We discussed with them what potential changes they believed would improve the Postal Museum's pricing structure, as well as who else they believed we should consult at the museum or elsewhere. Interviews were conducted instead of surveys, as we wanted to ask open-ended questions that generate well-thought-out answers that only can occur from a non-standardized exchange that the interviewee is already prepared for. These interviews were somewhat informal, as we were doing an internal review, but we still took notes in these interviews to be analyzed for any unique system changes, as well as a statistic analysis of general staff's feelings about the current system.

3.1.2. Benchmarking Against Other Museums

We analyzed other museums' pricings of related items. To do this, we first examined the websites of museums and archives to determine their pricing structures for different materials available for reproduction. These organizations included Image Leicestershire, the London Archive, Britain from Above, Historic England, and the Society of Antiquaries of London. Additionally, to enable more comprehensive research, employee contacts of several other organizations that offered similar licensing services were received from TPM staff. To improve our understanding of these organizations' pricing, we reached out by email to these employees to host informal interviews. These interviews were in person, if possible, but were limited to online meetings and email if appropriate. They provided us with the rationale behind their pricing scheme and any details that simply reviewing a website, without placing an order, may miss. In addition to pricing, general practices regarding image licensing were also discussed (see section 3.2.1).

Meanwhile, we had to discuss the new pricing system with TPM copyright staff, to avoid breaking any copyright and licensing laws, as well as adding copyright fees to situations that have had relevant laws established since 2019.

3.2. Objective 2 - Assess Current and Desired System Functionality

In order to improve TPM's current Image licensing process, we needed to look deeper into the current practices of the Museum to determine what changes were to be made. The first way this was accomplished was by conducting research into the DAMS technologies used by TPM, including PhotoShelter and the online catalogue, and how they could be used more effectively in the licensing process. The second way we examined this was by meeting with the employees of TPM responsible for image licensing and its related processes. From these meetings, our team aimed to gain a better vision of the project's scope and determine the problems that should be targeted for improvement.

3.2.1. Assess DAMS Functionality

Research was conducted into the two DAMS systems used by TPM in the image licensing process: PhotoShelter and the Museum's online catalogue. This research aimed to gather a better picture of how these two systems were used in the licensing process and what functions they could offer to the Museum that could improve current processes.

PhotoShelter was evaluated to gain a comprehensive understanding of its functionality, suitability for image licensing, and current use within TPM. To conduct this evaluation, we:

- Attended online meetings with PhotoShelter staff members Anna Hastings, Customer Success Manager, and Guy Johnson, Onboarding Team Manager, to learn Photoshelter's functionality.
- Reviewed PhotoShelter's website, documentation, and other online resources to understand details of its features and capabilities.
- Consulted TPM staff to understand why the platform was selected, how it is currently used, and to identify gaps in functionality.
- Consulted external organizations to understand other processes and platforms.
- Considered areas where complementary tools or integrations could enhance the platform's effectiveness.

The online catalogue was investigated to determine its role in the image licensing process. In our meetings with staff of TPM, we sought to better understand the involvement of the catalogue in managing images, handling day-to-day licensing enquiries, the rationale behind its current design and usage, and how it could be used to streamline existing image distribution workflows.

As PhotoShelter and the online catalogue were already known to be integrated into TPM's operations, the ultimate objective of this research was to qualify the Museum's utilization of these DAMS and identify ways to improve their value. To achieve this, we discussed image licensing practices with our organizational contacts in London (see section 3.1.2). These meetings assisted in identifying limitations and areas for improvement in TPM's current practices by exploring common challenges experienced when licensing items. These insights provided valuable context on issues such as: asset discovery, user access, content distribution, and e-commerce functionality. These findings were then compared to TPM's requirements and current processes to inform recommendations for creating an e-commerce platform in PhotoShelter and enhancing the effectiveness of the existing process.

3.2.2. Identify Targets For Process Improvement

Our team aimed to streamline TPM's image distribution process to conserve Museum resources, increase the Museum's capacity to process enquiries, and reduce friction for clients when they request images. To identify potential targets for improvement in TPM's image distribution process, it was first crucial to first gain a deeper understanding of current practices. Our team held weekly meetings with Museum employees who most directly managed the licensing process: Gavin McGuffie, Louis Porter, and Chris Taft. We were referred to other employees when more specialized knowledge was needed, such as financial integration. These meetings reviewed crucial details about TPM's distribution practices, including the volume and nature of enquiries received, how images are created, stored, and located, and staff's vision for the overarching role of the image licensing service. We developed a preliminary set of topics to discuss (see Appendix A) regarding the goals, methods, and desired functionality of the Museum's image licensing process. These questions were used to further our knowledge of these topics and served as a starting point for our investigations.

To determine which aspects of the distribution process should be targeted for improvement, we reviewed issues that were being faced by Mr. McGuffie, Mr. Porter, and Mr. Taft. Using this information, we were able to identify several key points of contention that could be remedied within the scope of our project.

3.3. Objective 3 – Design a More Streamlined Distribution Process

Our fourth objective was to prototype changes to the public image distribution portal in PhotoShelter, as well as to design the image distribution process to be more streamlined based on the findings from the previous objectives. Looking back to the goal of improving the online image distribution process for TPM, this objective served to meet that grand mission by delivering the basic framework portal and an improvised system flow.

The core objective of this project was to enable the public to purchase available images directly, so our priority was to enable and configure the public e-commerce portal on PhotoShelter. We conducted a thorough investigation of TPM's PhotoShelter account and assessed the system. To fully understand PhotoShelter, we also consulted with the PhotoShelter support team and received onboarding training for the system. The onboarding team answered our questions and explained the technical details. After we gained sufficient knowledge from them, we started testing the settings in PhotoShelter. In the process, we formulated a comprehensive guide to the portal for TPM to follow and configure the portal in the future.

After we had a feature plan for the portal, we moved to streamline the image distribution process by implementing new tools and investigating additional features. In the first few weeks, we discussed and investigated issues with the current process and proposed various solutions. For easy identification of the quality of images in the library for TPM staff, we found a new way to zoom in on images and requested a native zoom feature from PhotoShelter. We also wrote an external program to check the quality and flagged unqualified images. To ensure buyers can find the images they want, we investigated the searchability of images in TPM's online catalogue website and its PhotoShelter library. In the process, we discovered the main issue to be the inconsistency of image metadata between the catalogue and PhotoShelter library. We attempted to find ways to mitigate the issue, and we ultimately recommended TPM to gradually correct the metadata correspondence in the future. To get the buyers also informed and prevent

constant back-and-forth communications, we wrote a comprehensive guide page that includes detailed instructions on image licensing and using PhotoShelter.

3.4. Objective 4 - Implement the Prototyped System

The final objective of the project was to implement the system. While the physical implementation of a live commercial-ready distribution portal was not within the project's scope, we presented a working proof of concept to TPM staff and collaboratively developed a plan for implementation. Our team's role was to demonstrate what the system could look like and how it would function, and to work with TPM stakeholders to identify the steps needed to bring it into operation, and not to make those decisions individually.

4.0 Findings

This chapter presents the preliminary findings on The Postal Museum's image licensing process and the tools that could make that process easier to manage. The findings are organized around the path an image buyer would likely take: finding an image, understanding whether it can be licensed, completing payment, and receiving the final file. This structure helps show how the online catalogue, PhotoShelter, image quality review, metadata, and pricing all connect as part of one larger process.

The findings show that many of the tools needed to improve this workflow are already available to The Postal Museum. The online catalogue can be used as a starting point for users looking for images, while PhotoShelter can support many of the technical steps involved in selling, licensing, and delivering those images. However, the process depends on how clearly these tools are connected. Users need to know where to begin and where to go next, while staff need reliable metadata, clear image quality review procedures, and pricing categories that are manageable and easy to explain.

PhotoShelter can support much of the licensing workflow, but it works best when users can move through the system easily and staff can still step in where judgment is needed. With a clearer process, routine image requests can take less time, while more complex questions about copyright, pricing, image quality, and access can still be handled carefully.

4.1. Current Image Request System

The current process for handling client requests for image licenses is demonstrated in Figure 6 below:

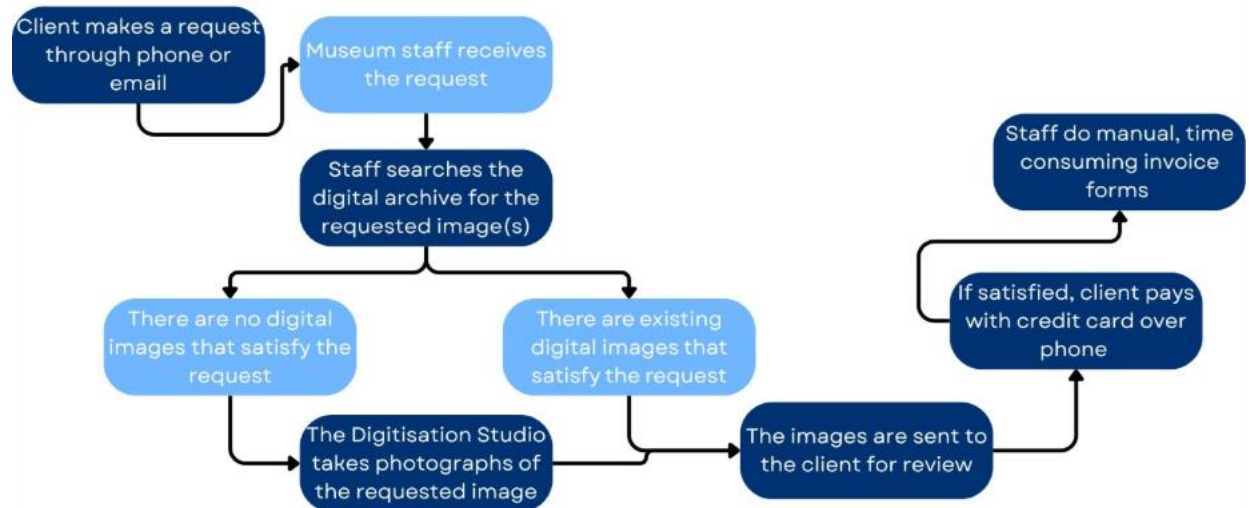


Figure 6: Current public image distribution system of the Postal Museum

This process starts when the Museum receives an enquiry from a client, who may discover the museum's image licensing service from many different sources. Some clients may have already browsed the catalogue and know the name of the images they want; many clients have much less information about what they need before contacting the museum. Once an enquiry is received, an employee must browse PhotoShelter to locate an image(s) that is suitable to fulfill the client's request and ensure that the image(s) are of sufficient quality. If the client desires an image that has not yet been digitized, an additional fee will be incurred for the item to be digitized in the Museum's digitization studio. The employee must also assess each image's copyright status to determine any applicable usage restrictions or licensing fees. Once all pertinent information is gathered, the employee works with the client to decide on a fee for the image. The employee typically proposes the fee based on a pricing spreadsheet; however, the true fee can vary widely depending on the nature of the enquiry. While fees for students, researchers, and most non-commercial use are nominal, fees for established organizations or large-scale commercial use are substantially higher. Once the fee is agreed upon, the client is

sent a link to the image they are purchasing and is informed that they will receive an invoice from TPM. Payment is completed over the phone or email, and the Museum's finance team completes the invoice manually.

As previously mentioned, this complexity in pricing is furthered by the legal considerations surrounding image rights. When handling image requests, the museum must navigate the copyright status of each individual image. Every item requires individual assessment, with staff determining ownership, permitted uses, and any restrictions on reproduction or distribution (personal communication). This process often involves consulting internal records, rights documentation, or potentially external sources, which can be time-consuming and not always clearly documented. Moreover, different types of usage (such as commercial, academic, or personal) carry distinct licensing conditions and fees, adding another layer of variability to pricing decisions. As a result, the inability to standardize copyright evaluation limits opportunities for automation and increases reliance on staff expertise and manual judgment, placing further strain on museum resources.

TPM uses two DAMS for the management of its images: the online catalogue and PhotoShelter. The online catalogue serves as a public-facing resource as opposed to PhotoShelter, which, prior to our project, was only used for backend image management. The catalogue is browsed primarily through keyword searches and sidebar filters (Using the Catalogue, TPM). Most records in the catalogue only consist of record descriptions and not the images themselves for security reasons - only the Museum's digital-born imagery and certain digitized material are viewable in the catalogue (The Postal Museum, n.d.). Ideally, the catalogue serves as an entry point for clients to locate records of images they want prior to submitting an enquiry. However, for several reasons, many clients simply contact the Museum first.

Several opportunities for improvement were identified in our review of this process, including implementing image flagging or tagging systems to simplify rights management, reducing the number of manual invoices generated, minimizing back-and-forth email communication through greater automation and self-service functionality, and introducing a clearer, and more structured pricing model, utilizing PhotoShelter's individual tiers.

4.2. How is PhotoShelter being used and what can it be used to do

Our review found that PhotoShelter enables users to search for assets using keywords, phrases, and filenames, supported by filters such as file type, orientation, date, and other attributes. The platform also supports custom metadata, including the ability to distinguish between free and paid images. Through the portal, users can discover, request access to, and download content, making it suitable for both internal and external audiences (PhotoShelter, 2024).

The evaluation identified several strengths. PhotoShelter was found to be **user-friendly**, **highly customizable**, and effective at **protecting digital assets**. It also offered strong **integration** capabilities with external platforms, including e-commerce systems, which aligned with TPM's objectives. Furthermore, the platform appeared **adaptable** to a wide range of sectors and user groups beyond photography, including government and healthcare organizations, which suits the museum.

The evaluation also identified several limitations. PhotoShelter can represent a significant financial investment, and some organizational requirements may require additional tools or integrations to be fully addressed. Since TPM already had the existing infrastructure for this program, this was less of a concern for our project. Due to this, we identified areas where supplementary systems may be beneficial, particularly in relation to e-commerce functionality.

While this evaluation examined PhotoShelter as a platform in general rather than its specific use at TPM, several of the identified strengths directly addressed challenges identified in the previous analysis of TPM's existing image request and sales processes. The analysis of TPM highlighted issues relating to image discoverability, customer access to collections, and the lack of a streamlined system for managing image requests and sales. As PhotoShelter already served as TPM's primary DAMS, its search functionality, metadata capabilities, and public-facing features presented an opportunity to address these challenges without requiring the implementation of an entirely new system.

Previously, PhotoShelter only functioned as a backend DAMS for TPM. While there was public-facing potential, it was not being utilized by TPM. This limited the museum's ability to efficiently process customer requests and promote image sales despite already maintaining its image collection within PhotoShelter and paying for its services.

We created a public-facing Collection and Gallery for TPM to use in order to list images publicly for sale. This approach was selected because PhotoShelter already acted as the functioning digital archive for the museum's image assets and provided user-friendly tools for searching, organizing, and requesting content. PhotoShelter was built for the purpose of generating income on an online gallery. So, rather than introducing an additional platform, the project focused on extending TPM's existing infrastructure to create a more efficient process for both customers and staff.

In creating a sales system, we were faced with several obstacles, most notably online visibility, configuring payment systems, and creating pricing tiers. In order to work on image visibility, we worked on improving the metadata of backend images. This will enable PhotoShelter to fully utilize a feature that creates alt text from the metadata. Alt text helps online visibility and lets images show up in more search results, thus improving online visibility for TPM's public platform. Improving metadata also addressed issues identified during the analysis of TPM's existing collection management practices, as accurate metadata improves both internal asset discoverability and external search engine visibility.

For the payment systems, we had to consider what PhotoShelter supported, and we worked to determine the best option for TPM using information on transaction fees, VAT charges, and ease of use for consumers. Ultimately, we decided with the assistance of TPM to use Stripe as the primary method of payment on the site. The revised workflow enables customers to discover images through the public portal, submit requests, and complete purchases through an integrated system, reducing the amount of manual intervention previously required from museum staff. The following sections describe the development of the Public Portal, the enhancements made to the backend metadata structure, and the research that informed decisions relating to visibility, pricing, and payment processing.

4.3. How other museums handle pricing structures

There were many different pricing schemes of the 8 different organizations we researched. Each organization had differing price cut-offs, and items they focused on selling. Across the organizations we examined, there were a few similarities between them that the Postal Museum did not share. The strongest commonality was a significantly less complex and

sub-divided pricing scheme. Across all of them, they averaged 2 pricing tiers per item, while the Postal Museum averaged 4, and got up to 7 for a single type. For example, the Royal College of Physicians, which has the most similar item types as the Postal Museum, has only a few items with even 2 different tiers of cost. Table 1 clearly shows a significant difference in complexity. The other organizations are similar, with a focus on concise, clear, and efficient price estimations.

Table 1: Comparison between similar Postal Museum and Royal College of Physicians pricing schemes (not a complete list of the Royal College, as they have more product types to sell)

What the Image is Used for	TPM	Royal College of Physicians
Magazine	5 subcategories with 7 tiers each, based on publication number	2 subcategories with 2 tiers each, based on publication number
Website	3 subcategories with 3 tiers each, based on duration	2 subcategories with 1 price each, based on if website is paid or free
TV/Film	5 subcategories with 3 tiers each, based on duration	1 price
Exhibitions	3 subcategories with 3 tiers each, based on duration	2 subcategories with 1 price each, based on if permanent or temporary

Another commonality we identified was a policy of charging money for personal purchases. In the case of Image Leicestershire and the London Archive, a significant amount of their business comes from purchasers from individuals. The other organizations we analyzed have similar policies on charging customers purchasing images for personal research or other uses, ranging from five pounds up to twenty per image. TPM does not have such a policy, only charging if the intended usage exceeds a thousand publications.

4.4. Updated Pricing Structure

While benchmarking TPM's pricing against the other organizations, we found several product types that need prices adjustments. We also found some product types that we could not

compare to other museums due to notable differences in what they were selling. This occurred in the “Exhibition/Display” and “Website” pricing. The pricing tiers for TPM’s “Website” item did not compare to other organizations, as shown in Table 2. TPM adjusts its price based on whether the buyer is using it for academic, editorial, or promotional purposes, as well as licensing the image for a year, ten years, or forever. If we just examine the raw numbers found in Table 3, then TPM falls in the middle of the prices, except for the promotional prices, which are quite high.

Table 2: Tabulated summary of different museums’ pricing considerations when selling images to be used for websites.

Organization	TPM	Royal College of Physicians	Society of Antiquaries of London	Britain from Above
What they consider	What the website is being used for (Academic, Promotional, Editorial), Duration	Is it free to access the website, or does it need payment?	No consideration, one price for 5 years	No consideration, one price for 5 years

We could not compare TPM’s “Exhibition/Display” pricing with other organizations, as the license durations are 1 week, 6 months, and 10 years. These time periods fall far shorter than the Society of Antiquaries of London’s durations of 1 and 5 years, while the 10 years value is nearly identical in price to the Royal College of Physicians’s “Permanent” price. Like the “Website” prices, though, TPM’s raw numbers are comparable to the other two museums, and so likely do not need to change.

TPM’s prices for newspapers were underpriced when compared to the Royal College and Britain from Above but were significantly more expensive than the Society of Antiquaries of London, as found in Table 4. TPM, Britain from Above, and the Royal College divide their prices between the front cover of the newspaper, and inside the newspaper, with the latter being less expensive. The Royal College followed a similar pricing scheme to TPM, where the price of the image being inside the newspaper is half the cost of the front cover, while Britain from

Above set the inside section at a quarter of the front cover price. Britain from Above sets their front cover price relatively high, however, and so the difference in the organizations' pricing is as great as it would be otherwise. The Society of Antiquaries of London did not divide their pricing, simply setting the price of having their image in a newspaper at a comparatively low price of 40 pounds.

TPM's "TV/Film" pricing was the most difficult to properly benchmark, as TPM has the unusual policy of allowing five inserts of an image, per charge, for a duration. As no other museum or business does something like this, we both compared the values as they are, and divided the TPM prices by five to compare to other single image prices. With this adjustment, we found that the prices were significantly lower than any other organization, while the unadjusted prices were on the higher end (Seen in **Table 5**). Historic England, the Royal College, and the Society of Antiquaries of London all had similar durations to TPM but were priced at a greatly increased value when compared to the divided price. The Society of Antiquaries of London was priced at nearly the same cost as one of TPM's tiers, but only for a single image.

We also reached out to the Royal Horticultural Society, which used to have a very similar business to TPM. Over the years, they found through financial review that the business was not producing more money than it took to keep running and so shifted their focus to selling posters or aesthetics to individuals. The new business venture has been properly working for a few years now and is selling much better than the old business practice, which could be a potential additional venture for TPM.

4.5. Streamline Process

After setting up the public portal, we then turned to improve the user experience for both the TPM staff and customers using the PhotoShelter portal by streamlining the image distribution process. In the first few meetings, we heard a few grievances from the TPM staff about the system. We also conducted investigations on our own to find potential problems. We identified three major concerns – image quality evaluation, image searchability, and information availability. All three issues were addressed with extra functionality or recommendations.

4.6. Image Quality Evaluation

From photographer Louis, we received the issue of the lack of an easy method to check the quality of images in the PhotoShelter library for TPM staff. Some images in the library do not meet the quality standards to be licensed, so TPM would like to examine the quality of images before making them publicly available. The first part of the problem was that the images cannot be zoomed in, so the staff cannot check them unless they download the image. The second part was that there was no automatic process for evaluating image quality, and manually checking tens of thousands of images and picking out the problematic images is very time-consuming. To address the zoom in function, we discovered laptops and touchscreens support quick web browser zoom in function and recommended it to Louis. We also requested a native zoom-in feature from PhotoShelter to be developed. For the second problem, we built an automated process to evaluate image quality with a program named Image Quality Examiner (IQE). IQE can search through any gallery and collection in the PhotoShelter library to find images that need to be reviewed based on its metadata. Then it can tag images by attaching keywords to those images and showing reasons for review. TPM staff can search for those keywords in the library and find the problematic images all in one place. A more detailed IQE process is illustrated in figure attached below.

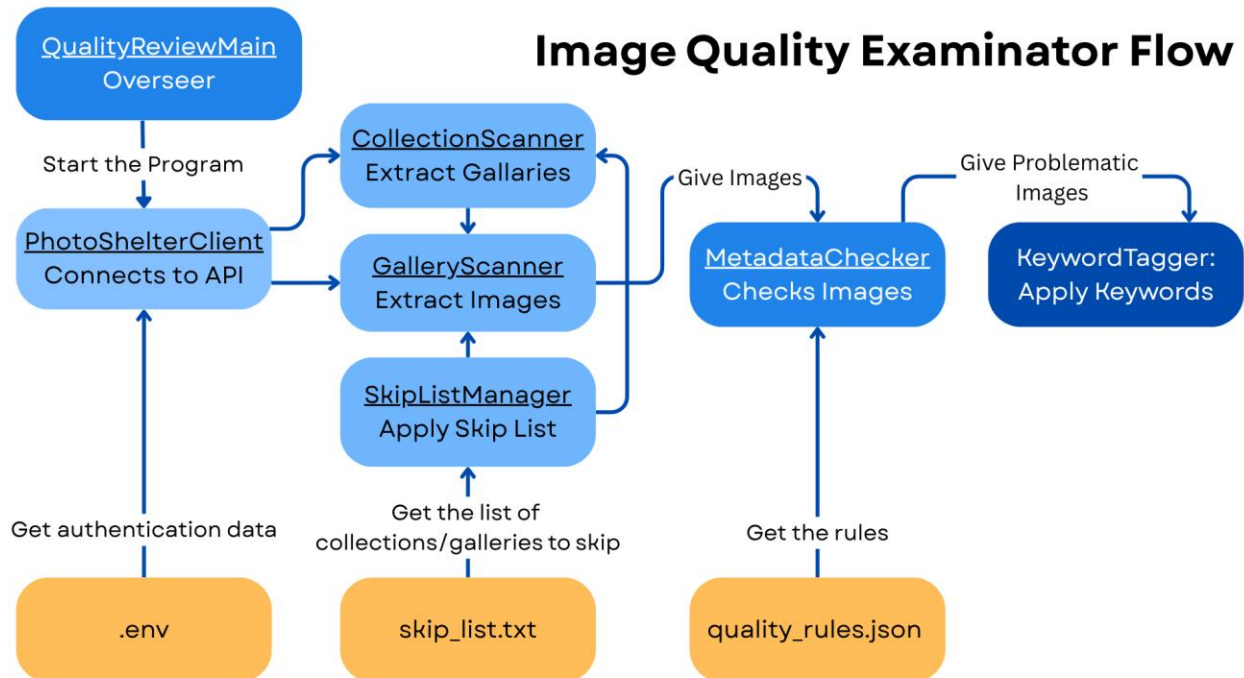


Figure 7: The flow chart of the Image Quality Examiner. It starts at the top left with the *QualityReviewMain* file to act as the overseer of the whole process. The process begins with *PhotoShelterClient* connecting to the *PhotoShelter* API and getting access to server data. Once connected, the scanners extract all the images under the collection or gallery the user requested to scan. In the process, some collections and galleries can be chosen to be skipped by the user. Then all the images are sent to the *MetadataChecker* and checked against the metadata rules. Finally, it sends a list of problematic images to the *KeywordTagger*, which tags those images by uploading the appropriate keywords to the *PhotoShelter* library.

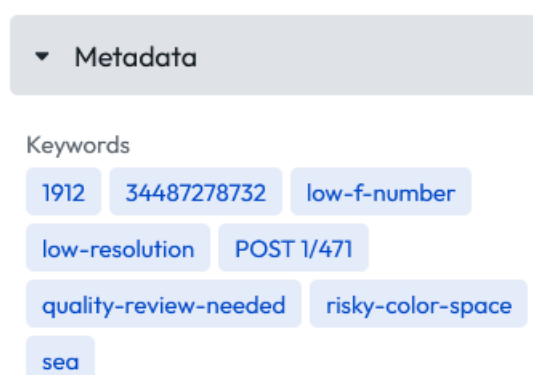


Figure 8: The example keywords of an image after being scanned and tagged by the Image Quality Examiner (IQE). Every problematic image gets the keyword 'quality-review-needed', and the keywords corresponding to its problems, like 'low-resolution' and 'low-f-number'. These keywords make problematic images very easy to search and filter for TPM staff.

With this program, the process of evaluating is significantly quicker and easier. However, there are major limitations. IQE only evaluates an image based on its metadata, so images with incomplete metadata or visual errors can go under the radar. The program also requires

significant computation power and time to examine all the tens of thousands of images in the library. Despite these limitations, IQE still received positive reviews from the staff and can be further developed by TPM in the future. Deliverables of the program include the source code along with its documentation (Appendix C).

4.7. Image Searchability

In the second week of the project, we learned that the online catalogue is currently the primary platform for customers to find their desired images and items to digitize. We investigated the online catalogue and determined that it is a viable starting platform for the image distribution process (Appendix D). It has adequate searching and browsing functionalities, and every item and image in TPM's collections can be found there. Therefore, we decided to keep the catalogue as the first browsing website for photo buyers. In the new process, they will be guided to the PhotoShelter portal to search and purchase that image. But it also has a major issue - the inconsistency between the metadata of the image in the online catalogue and in the PhotoShelter library. For example, the figures below show an image from the catalogue having the previous number 'H 12955/3', but that image's name is of a different format as 'H12955_003' in the PhotoShelter library. Combined with the limited keywords and description metadata of the image in PhotoShelter, this means no quick way to find that image in PhotoShelter using the catalogue information.

Still shot from 'Night Mail' (1936)

[« Back to search results](#)

For specific questions you would like
answering please email us :
archives@postalmuseum.org

For ordering and viewing options, select Ordering and Viewing

[Please sign in to your account](#)

Details	Browse by Hierarchy	Ordering and Viewing	User Comments
Held at:GB 813 The Royal Mail Archive, Calthorpe House Finding Number:POST 118/17611 Previous Number:H 12955/3 Date:1987 Level:File Extent:23 film negatives, 10 film positives Format:Photograph Description:Photograph. Image orientation: Monochrome. various. Reproduction made 1987 from a 35mm print of the GPO Film Unit film 'Night Mail'. Features a train.  Add to Wishlist ▶			

Figure 9: An image in TPM's online catalogue. It has various metadata, like name, finding number, previous number, and description.

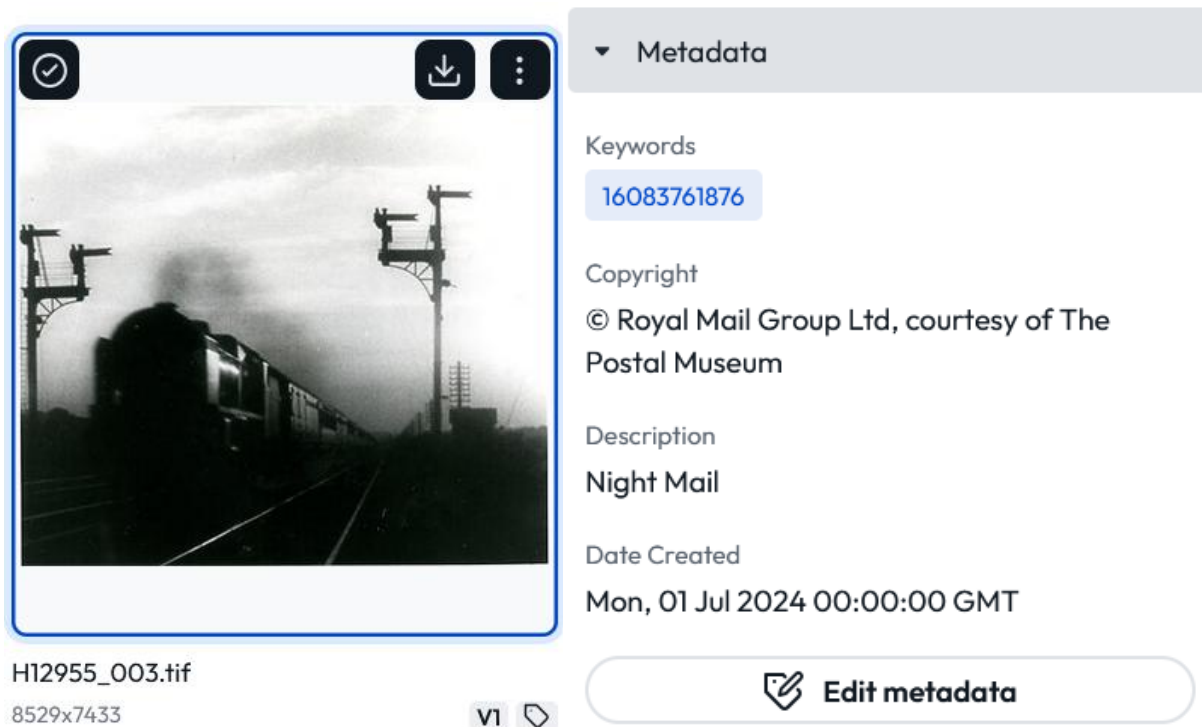


Figure 10 (Left): The same image from Figure 9 but in PhotoShelter. Notice that its name is the previous number in a different format.

Figure 11 (Right): The metadata of the image from Figure 10. The keywords field and the description field are lacking, making it difficult to find.

Some images are even in different sections, like an image is in 'POST 20/415' section of the catalogue, but of 'POST 119/177' in the PhotoShelter library. In conclusion, these inconsistencies will make it difficult or impossible for the customers to find some images in the PhotoShelter library using the information in the catalogue. To solve this, we looked into ways to correct the metadata correspondence between the catalogue and the PhotoShelter library, as well as functions for users to search images in PhotoShelter without detailed manual metadata.

The first search did not bring exiting new discoveries. After sampling some images, we predict the majority of the images are searchable in the PhotoShelter library, either they have a correct labeling number format, or there are other metadata and functions that help the search. But there are problematic images scattered throughout, and we did not find an easy way to

correct these metadata correspondences. We recommend that although it is not a very urgent issue, TPM should consider gradually improving consistency manually in the future.

The second search generated some solutions, but none are ideal. One, images have automated metadata generated by PhotoShelter. Those keywords are very generic, like ‘Book’, ‘Wheel’, and ‘Poster’. They can assist in the search for images, but the search often returns a large number of results and makes it difficult to find the right one. An example of automated keywords of an image is shown below.

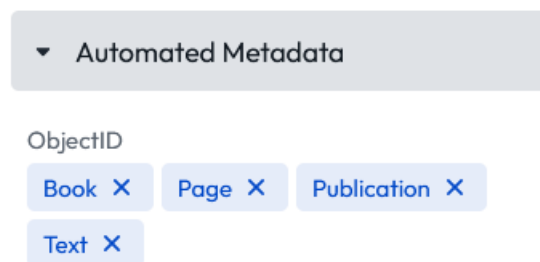


Figure 12: The automated metadata generated by PhotoShelter of an image. None of the keywords give specific details about the image, making them not much help in finding the image.

The second function is AI visual search, which uses AI to find images based on visual clues. It does return images related to search keywords, but it can take extensive time for the search to be completed. The AI works well for searching objects like ‘train’ or ‘mailbox’, but it performs much worse when it comes to finding documents, which is a big part of TPM’s collections. The third feature is Alt text, which is an AI generated description of the image. This could be very accurate, but manual checking is still needed to ensure the description is totally correct, portrayed in the two figures below.



Figure 13 (Left): A new uploaded image to PhotoShelter. It is the cover image of the user guide in TPM's online catalogue.

Figure 14 (Right): Alt text generated by PhotoShelter AI of the image in Figure 13. It describes the scene accurately but mistakes the text to be the label of the room.

Also, Alt text currently can only be generated on new uploads, not retroactively. So, the tens of thousands of images already in the library cannot have Alt text without reuploading them. In the end, all three features have major drawbacks and do not offer an ideal solution for the searching problem. Again, we recommended TPM to correct image metadata between the catalogue and library in the future.

4.8. Information Availability

Information availability to the customers was another area of concern for TPM. In the past, the only communication channel between customers and TPM staff was through email. And those emails spiraled into exhausting email chains of short back-and-forth exchanges. The source of the problem was that there was hardly any public information on the image licensing process, so customers had to ask the staff to know anything. Our solution is a detailed user guide page in the PhotoShelter portal for everyone to see (Appendix E). The guide introduces buyers to the catalogue and assists buyers with a step-by-step procedure to search for images in the PhotoShelter portal. It also links to existing guidelines on the TPM's website on commercial licensing and informs the communication channel with the TPM staff. Since the PhotoShelter portal does not support web design and hosting extra webpages, we instead gave TPM a word document template of the user guide page. We recommended that TPM edit the content, host the content of the user guide on its own website, and link it to the portal. By making this information

available to the public, we hope to streamline the communication flow between staff and customers. Ideally, customers will be able to largely guide themselves through the image purchasing process.

4.9. The Final System

After we addressed the three concerns above, we have a more streamlined image distribution process, for both TPM staff and clients.

TPM Perspective - With a public portal to automatically sell images, the image licensing business will become a more passive income source. The staff will no longer have to directly interact with every simple request, and the new user guide page should reduce back-and-forth communication with clients. They will also have an easier time conducting image quality evaluation with our Image Quality Examiner software. All of these improvements mean less workload and possibly a higher image licensing volume in the future.

Client Perspective – An online portal can be very convenient for clients. They can be deterred by the need to contact TPM staff, but the new process resembles online shopping, a very familiar experience for many people. Shown in Figure 17, the simplified ideal process is that the client finds the image they want on the online catalogue, goes on to the PhotoShelter portal, finds it, and purchases it directly. They may also browse other images in the portal in the process and discover other images they wish to license. The client will not need to email TPM and can have the images within minutes. If they get stuck during the process, the user guide page can provide the necessary information to help them.

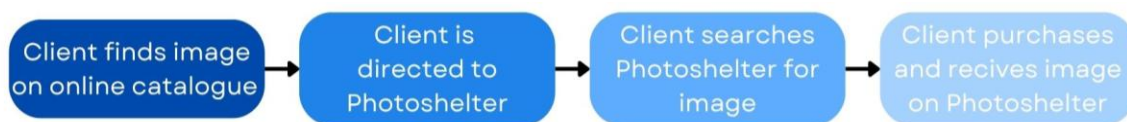


Figure 15: The simplified ideal image distribution process from a client's perspective. It is a straightforward process that can take less than 30 minutes to complete.

4.10. Presentation Of Final System

We presented a portal walkthrough demonstrating the system from two perspectives: that of a Postal Museum administrator managing the archive and approving image requests, and that of an end user, a journalist, publisher, or member of the public, navigating the portal to search for, license, and download an image. By showing both sides of the system, the presentation allowed staff to understand the public facing platform and address questions and concerns, as well as providing a way to familiarize staff with the changes.

This proof of concept was presented to key TPM stakeholders, including members of the Archive and Collections team, and a representative from the finance and IT teams. These are the staff most directly affected by a change to the image supply process. Rather than prescribing a fixed launch schedule, we worked to create a foundation for implementation and with TPM we developed a plan that reflects their capacity, infrastructure, and any outstanding decisions that fall outside the scope of this project.

5.0 Conclusion

At the start of this project, The Postal Museum already possessed many of the components necessary for an effective image distribution system, including a substantial digital archive, an established licensing process, staff expertise, and the use of PhotoShelter as a digital asset management platform. However, the existing process relied heavily on manual communication, individualized pricing decisions, and case-by-case copyright review, creating inefficiencies and increasing staff workload. This project sought to identify opportunities to improve the accessibility, efficiency, and sustainability of the museum's image distribution process.

To address these challenges, a revised image pricing structure was developed through internal evaluation and external benchmarking. The capabilities of PhotoShelter and alternative digital asset management approaches were assessed to identify opportunities for implementing a monetization system. Desired system functionality was established through analysis of existing workflows and feedback gathered from both staff and clients. These findings informed the design

and development of a prototype online image distribution portal intended to streamline and automate key aspects of the licensing process.

The resulting proof-of-concept system demonstrates how The Postal Museum could improve access to its image collection while reducing administrative burden on staff. Accompanied by a implementation plan, the recommendations presented in this report provide a practical foundation for future development. By building upon the work completed during this project, The Postal Museum can continue to expand access to its collections, enhance the user experience for image licensing clients, and strengthen the long-term sustainability of its digital image distribution program.

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Appendix A

Preliminary Discussion Topics with The Postal Museum Employees

Below is a tentative list of topics we plan to discuss with pertinent employees of TPM. We aim to have these conversations to better understand the current goals, practices, constraints, and desired functionality of the Museum's image licensing process.

- Measures the Museum takes to protect copyrighted images in the collection
- What determines the visibility of images in the catalogue
- Barriers that exist in publicizing image licensing and supplying prices to clients
- Barriers that exist in publicizing copyright, usage rights and other legal information associated with each image
- Volume of licensing requests the Museum receives, and has received in the past
- Methods by which image prices are determined
- Why TPM decided to use PhotoShelter as opposed to other services

Appendix B

Tables Comparing Prices Between Organizations

Table 3: Comparison of image licensing prices intended for use on a website

Table 3.a: The Postal Museum website prices (in pounds)

Website Type	Placement of Image	1 year	10 years	Unlimited
Academic	Homepage	50	70	90
	Inside Page	30	50	70
	Banner	20	40	60
Editorial	Homepage	110	130	150
	Inside Page	90	110	130
	Banner	80	100	120
Promotional	Homepage	500	550	600
	Inside Page	400	450	500
	Banner	350	400	450

Table 3.b: Comparable organizations' website prices (in pounds)

Organization	Duration	Price	
Royal College of Physicians	No time limit	150 if website is pay to access	24 if website is free to access
Society of Antiquaries of London	5 years	90	
Britain from Above	5 years	50	

Table 4: Comparison of image licensing prices intended for use in a newspaper

Table 4.a: The Postal Museum licensing prices (in pounds) by range of publishing distance and placement of image.

The Postal Museum		
	Local Publishing	National Publishing
Front Cover	100	240
Inside of Newspaper	50	120

Table 4.b: Britain from Above licensing prices (in pounds) by range of publishing distance and placement of image.

Britain from Above		
	Local Publishing	National Publishing
Front Cover	150	250
Inside of Newspaper	35	75

Table 4.c: Royal College of Physicians licensing prices (in pounds) by expected number of copies and placement of image.

Royal College of Physicians		
	Less than 1000 Copies	Over 1000 Copies
Front Cover	300	360
Inside of Newspaper	150	210

Table 5: Comparison of image licensing prices intended for use in a TV show or film

Table 5.a: The Postal Museum licensing prices (in pounds) by duration of use and range of transmission for 5 inserts

The Postal Museum (per 5 inserts)			
	1 Transmission	10 Years Unlimited Use	Unlimited Time and Use
Local	190	450	500
Single Country	240	500	550
Europe	240	500	550
North America Only	240	500	550
Worldwide	390	650	700

Table 5.b: The Postal Museum licensing prices (in pounds) by duration of use and range of transmission per single image

The Postal Museum (single image)			
	1 Transmission	10 Years Unlimited Use	Unlimited Time and Use
Local	38	90	100
Single Country	48	100	110
Europe	48	100	110
North America Only	48	100	110

Worldwide	78	130	140
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Table 5.c: Other organizations' licensing prices (in pounds) by duration and unique modifiers

Organization	Specifics	Price	
Historic England	Transmission range, scales by duration	One Country	96 - 288
		World	144 - 432
Society of Antiquaries of London	10-year license only	450	
Royal College of Physicians	Permanent Use	360	

Appendix C

Image Quality Examiner Document

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Date: June 17, 2026

Purpose

Image Quality Examiner (IQE) is a program developed by the 2026 WPI Team for The Postal Museum (TPM) in London. It aims to assist with quality assessment of TPM's images in the PhotoShelter library. This document outlines the technical details of IQE and instructions on using the program.

Main Functionality

IQE can search through any gallery and collection in the PhotoShelter library to find images that need to be reviewed based on its metadata. Then it can tag images by attaching keywords to those images and showing reasons for review. Those keywords are very searchable in PhotoShelter, and they can be easily removed after the image has been reviewed.

Specification

Language	Python 3.14
Integrated Development Environment (IDE)	PyCharm 2025.3.3
Files	QualityReviewMain.py PhotoShelterClient.py CollectionScanner.py GalleryScanner.py MetadataChecker.py quality_rules.json KeywordTagger.py .env skip_list.txt SkipListManager.py

File Review

File Name	Function
QualityReviewMain.py	This is the main orchestrator of IQE. It brings every component together and executes the program. The file hosts the user interface for using the program.
PhotoShelterClient.py	It handles connections and communication with the PhotoShelter API. It authenticates the program, sends requests, and manages the rate limit. Authentication setup in the .env file is required.
CollectionScanner.py	It extracts all galleries from a given collection, then use the gallery scanner to extract all the images. It also includes functions that extract galleries from a collection. No setup is needed.
GalleryScanner.py	It extracts all images from a given gallery. No setup is needed.
MetadataChecker.py	It receives a list of images from the scanners. It checks the images for quality violations and flags them. It loads the metadata checks from quality_rules.json. Ensure quality_rules.json is loaded. To add new rules, modification to the function _apply_rule is needed.
quality_rules.json	It contains all of the metadata rule checks. Modify the “enabled” value to enable/disable specific rules. Reference the existing rule structure when adding new rules.
KeywordTagger.py	From MetadataChecker, it receives a list of images along with their violations. Based on the list, it uploads the violation keywords of each image to the PhotoShelter library.

.env	It stores sensitive authentication information. If the .env file does not exist, create one with the following template: Template: API_KEY=[TPM'S API KEY] EMAIL=[TPM ACCOUNT] PASSWORD=[PASSWORD OF THE TPM ACCOUNT] ORG_ID=[TPM'S ORGANIZATION ID ON PHOTOSHELTER] Example: API_KEY=12FyU EMAIL=tpm@gmail.com PASSWORD=password ORG_ID=uIj62 If you do not know the API key, ask the WPI team or Louis.
skip_list.txt	This is a list of collections and galleries to skip when doing image quality assessment. Users may add and remove names of the collections and galleries for their images to not be checked. Users may choose to apply or ignore the skip list in QualityReviewMain.
SkipListManager.py	It manages the skip list. It loads the list from the file and tells the scanners whether to skip this collection or gallery.

How to run the IQE?

1. Download the files above. Ask Louis on how to get them.
2. Install Python 3.14 on the computer. The installation link is <https://www.python.org/downloads/release/python-3145/>
3. Recommend using an IDE to run the program. This program was developed on PyCharm: <https://www.jetbrains.com/pycharm/>. Otherwise, run it in an IDE of your choice. Visual Studio Code was confirmed to be a working platform.
4. Check for issues while opening the folder in the IDE. Common tasks include installing necessary packages, setting up a Python 3.14 interpreter, and creating the .env file.
5. Also recommend looking around and seeing the comments on the files.
6. To execute the program, go to QualityReviewMain.py and scroll to the bottom for the main function. There, you can configure the run setting like collection/gallery name, mode, dry run, and apply skip list.
7. Once you configure the run, execute QualityReviewMain.py. While it is running, check the console messages. They are progress updates and can tell you what's wrong.

Examine Images on PhotoShelter

When a run is complete and keywords are uploaded, you can see the effect immediately. Go on PhotoShelter library to examine the images. Some things to consider:

1. Every tagged image will have the keyword “quality-review-needed”. On top of that, each image will also have specific keywords like “low-resolution” or “auto-exposure” to show what metadata rule it failed.
2. You may search the keywords directly to see all the images that have them. For example, search “quality-review-needed” to see every image that needs to be reviewed.
3. Keywords can be easily removed by editing metadata on the website.

Appendix D

TPM Online Catalogue Analysis

The Postal Museum's Online Catalogue: <https://catalogue.postalmuseum.org>

Online Catalogue's Relation to the Image Licensing Business:

The online catalogue is the entry point for potential image buyers. They will browse and search the catalogue for images they wish to license from TPM. When the buyers find the images, they will be transferred to PhotoShelter to complete the transaction.

Analysis Conclusion:

The online catalogue is a viable entry point for image buyers. Users can browse and search for catalogue items efficiently. Directing users from the catalogue to PhotoShelter should not be difficult. Finding the corresponding image in PhotoShelter from existing imaged items in the catalogue is largely fine. Some updates to the PhotoShelter library might be necessary.

Tasks:

1. Evaluate the browsing and searching features of the online catalogue.
 - What is the experience of browsing the catalogue by category?
 - What is the experience of searching for specific images?
2. Identify potential methods to transfer buyers to PhotoShelter.
 - How to get the buyers to go on PhotoShelter smoothly?
3. Evaluate the correlation between catalogue and PhotoShelter.
 - How could the buyers find the images on PhotoShelter based on the catalogue?
 - How much is the PhotoShelter library equal to the catalogue?

Task 1: Evaluate the browsing and searching functionalities of the online catalogue.

Feature	Current Status	Conclusion
Browsing	The catalogue has an instruction page called “Using the Catalogue”. It contains the way the catalogue is organized and how to use the sidebar. The side bar does narrow the results with various categories, like Subject and Format. The categories lessen or expand according to the current search. The page refreshes with every sidebar update. The images tab displays many searched images at once. The map tab displays a few images on the UK map. The timeline is blank for me.	The browse feature is working. The images tab is good for visual search. Frequent page fresh, map and timeline tab could be improved, but they do not influence the core function. Buyers should not have major issues browsing the catalogue.
Searching	“Using the Catalogue” also has detailed instructions for searching. The catalogue supports advanced searching and wildcard characters. The search does return results containing the keywords in the name or description. Advanced search and wildcard characters do work. Most items in the catalogue do not have an associated image. There is an ‘image only’ tab at the side bar that only displays items with an image.	The search feature is working. Items of associated keywords do show up. Most items in the catalogue do not have images, since only a small portion of the collection is digitalized. More preview images could attract more buyers, but that requires increased digitalization effort.

Task 2: Identify potential methods to transfer buyers to PhotoShelter.

The easiest way to transfer the buyers is to include the PhotoShelter homepage link on the catalogue page. An additional link of every item to its corresponding image on PhotoShelter is not viable in the current state, since most items have not been digitalized.

Task 3: Evaluate the correlation between catalogue and PhotoShelter.

Searching:

Searching for the catalogue number of an imaged item usually works in PhotoShelter.

Some images on PhotoShelter are difficult to search for. For example, there is a catalogue item with the name 'Still shot from 'Night Mail' (1936)'. Its Finding Number is 'POST 118/17611' and Previous Number is 'H 12955/3'. The corresponding image in Photoshelter has a name of 'H12955_003'. It does not have the finding number attached, and it does not have the correct previous number as a keyword. It also has a short description of 'Night Mail', so the search has 9363 results.

Mismatched Inventory:

Some of the items in the catalogue are not in PhotoShelter, which is fine since not everything is digitalized. Some items in PhotoShelter are not found in the catalogue, which is strange. And the numbering system might have errors.

For example, 'POST_119_177_inside' is a document image in PhotoShelter, but I could not find the item in the catalogue. The last number of the Post 199 series on the catalogue has four digits, like 'POST 119/1067'. POST 119/177 does not show up in the catalogue search. The catalogue item 'POST 20/415' has a similar description to 'POST_119_177_inside', but I also could not find 'POST_20_415' in PhotoShelter. This might be an error or an accident, but it is suspicious.

There are duplicate images in PhotoShelter. Under Post 188, there is a 'Night Mail' group and a 'Nightmail jpegs' group. Night Mail has a 'H12955_002.tif' with keyword 16083573664. Nightmail jpegs has a 'H12955_002.jpg' with keywords 16087086574 and H 12918/11. They are two separate images in the system, but they look the same. A potential issue is that duplicate images may confuse buyers.

Appendix E

User Guide Template for the PhotoShelter Portal

The Postal Museum's Online Catalogue Site: <https://catalogue.postalmuseum.org>

Our online catalogue contains more than 120,000 records describing the files, objects, stamps, posters, photographs, and other materials in our collections. You can read more about it at <https://www.postalmuseum.org/collections/catalogue/>. We recommend you **browse the catalogue first**. You may find interesting existing images or items that are not digitized that you want to get an image of. To license images, come back to the PhotoShelter portal and follow this guide.

Licensing

This portal is intended to license images for personal use and academic purposes. You may purchase images under the available usage option applicable to you. For commercial licenses, you need to submit a licensing request form to the museum's email. Please read more about it in <https://www.postalmuseum.org/collections/licensing/>.

Search Images using the Catalogue

If you see images in the catalogue that you want to acquire, most likely their high-quality versions are in the PhotoShelter library. To search images in the PhotoShelter portal, we recommend following the following steps:

1. Search the image by using the Previous Number or Finding Number of the image in the catalogue.
2. If the desired image does not appear, change the format of the Previous Number by removing spaces, replacing '/' with '_', and adding zeros. For example, 'H 12955/3' becomes 'H12955_003'. With the new Previous Number, search the portal again.
3. If the desired image does not appear, search for the image using other information. Try searching for the name, date, or keywords from the description of the image.
4. If the desired image does not appear, turn on the AI visual search by clicking on the green button to the right of the search bar. Try searching for the defining features of the image, like 'train' or 'watch'. The process usually takes some time, but you can start checking the results by double clicking on the first result, even if it is a gray square.

If you cannot find the desired images, please send a request to our staff by emailing archives@postalmuseum.org. Include the link of the image in the catalogue and/or its description in the email. We will get back to you as soon as possible.

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